

Control of NFAT Isoform Activation and NFAT-Dependent Gene Expression through Two Coincident and Spatially Segregated Intracellular Ca²⁺ Signals

SCIFACT-EVIDENCE-30303335

Evidence abstract

Excitation-transcription coupling, linking stimulation at the cell surface to changes in nuclear gene expression, is conserved throughout eukaryotes.

How closely related coexpressed transcription factors are differentially activated remains unclear.

Here, we show that two Ca²⁺-dependent transcription factor isoforms, NFAT1 and NFAT4, require distinct sub-cellular InsP3 and Ca²⁺ signals for physiologically sustained activation.

NFAT1 is stimulated by sub-plasmalemmal Ca²⁺ microdomains, whereas NFAT4 additionally requires Ca²⁺ mobilization from the inner nuclear envelope by nuclear InsP3 receptors.

NFAT1 is rephosphorylated (deactivated) more slowly than NFAT4 in both cytoplasm and nucleus, enabling a more prolonged activation phase.

Oscillations in cytoplasmic Ca²⁺, long considered the physiological form of Ca²⁺ signaling, play no role in activating either NFAT protein.

Instead, effective sustained physiological activation of NFAT4 is tightly linked to oscillations in nuclear Ca²⁺.

Our results show how gene expression can be controlled by coincident yet geographically distinct Ca²⁺ signals, generated by a freely diffusible InsP3 message.

Document metadata

Corpus | SciFact

Document ID | 30303335

Evidence checklist

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Source continuity

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